

PROJECT TOPIC: AIR POLLUTION & AIR QUALITY INDEX (AQI) **30-DAY MONITORING**

Topic Description: Empirical 30-Day Air Quality Monitoring Study Using Official AQI Mobile Applications (CPCB Sameer / SAFAR), Weekly Parameter Tracking (PM_{2.5}, PM₁₀, NO₂, O₃), Weather Correlation, and Health Advisory Synthesis

Ambient air pollution has escalated into the most severe environmental public health emergency in modern urban India, causing millions of premature deaths annually and triggering widespread chronic respiratory and cardiovascular illnesses. In the state of Maharashtra, rapid industrialization, massive construction booms, and exponential growth in personalized vehicular traffic have severely degraded air quality across urban centers—spanning Mumbai, Pune, Navi Mumbai, Thane, Nashik, and Nagpur. During winter months, meteorologically induced thermal temperature inversions trap toxic vehicle exhaust and industrial pollutants near ground level, creating dangerous toxic smog episodes.

To demystify complex atmospheric chemical concentrations into a clear, understandable public health metric, the Central Pollution Control Board (CPCB) launched the national **Air Quality Index (AQI)**. Operating on a color-coded numerical scale from 0 to 500 across six distinct categories (*Good, Satisfactory, Moderate, Poor, Very Poor, Severe*), the AQI integrates real-time concentrations of six primary criteria air pollutants: **Particulate Matter (PM_{2.5} and PM₁₀), Nitrogen Dioxide (NO₂), Sulfur Dioxide (SO₂), Carbon Monoxide (CO), and Ozone (O₃)**. Modern digital governance now makes real-time, satellite-calibrated monitoring data accessible to every citizen via official mobile applications such as *SAMEER (CPCB)* and *SAFAR (Ministry of Earth Sciences)*.

This project conducts a rigorous 30-day citizen science air pollution monitoring

2. Importance of the topic (विषयाचे महत्त्व)

Conducting a longitudinal air quality index monitoring study is of paramount clinical, environmental, and civic importance:

Crucial Reasons Why AQI Mobile Auditing is Essential :-

- **Unmasking the Invisible Killer (PM_{2.5}):** Fine particulate matter (PM_{2.5}, aerodynamic diameter $< 2.5 \mu\text{m}$) penetrates deep into human pulmonary alveoli and enters the bloodstream, directly triggering strokes, ischemic heart disease, lung cancer, and stunted childhood lung capacity.
- **Democratizing Environmental Science via Citizen Telemetry:** Transforms complex atmospheric chemical parts-per-million data into accessible digital knowledge, empowering students and citizens to track real-time health risks on their personal smartphones.
- **Identifying Local Urban Microclimatic Emission Hotspots:** Continuous monitoring pinpoints specific times of the day (e.g., peak evening traffic hours: 6 PM to 9 PM) and weather conditions that trigger acute pollution spikes, enabling targeted behavioral adjustments.
- **Holding Civic Authorities Accountable Under NCAP:** Generates empirical citizen records to verify whether municipal corporations are actively enforcing National Clean Air Programme (NCAP) mandates—such as dust suppression, construction enclosure, and mechanical road sweeping.

Clean air is an unalienable fundamental human right, and citizen monitoring is the first step toward securing it.

3. Objectives of project work (प्रकल्प कार्य उद्दिष्टे)

This 30-day digital air quality monitoring investigation was guided by structured empirical and public health objectives:

Structured Project Objectives :-

- **Digital Monitoring Protocol Setup:** To configure the official CPCB *SAMEER* and *SAFAR-India* mobile applications, linking data feeds to the continuous ambient air quality monitoring station (CAAQMS) nearest to our locality.
- **30-Day Longitudinal Data Logging:** To log composite AQI values and individual pollutant concentrations ($PM_{2.5}$, PM_{10} , NO_2 , SO_2 , O_3) at standardized times (8:00 AM, 2:00 PM, and 8:00 PM) every week for 4 consecutive weeks.
- **Prominent Pollutant Identification:** To determine the single dominant criteria pollutant driving the composite AQI score during each weekly observation cycle.
- **Meteorological & Traffic Correlation:** To correlate daily AQI swings with local weather conditions (wind speed, temperature inversion, relative humidity) and vehicular traffic density.
- **Health Advisory & Action Plan Formulation:** To synthesize a student-led Clean Air Charter proposing personal protective measures, anti-idling campaigns, and public transit adoption.

The study provides authentic empirical telemetry documenting urban atmospheric health.

4. Project Work Methodology (प्रकल्प कार्यपद्धती)

A structured digital telemetry tracking, statistical correlation, and environmental health assessment methodology was executed:

Methodological Phases Followed :-

- **Phase 1 – Mobile Application Calibration:** Installed the CPCB *SAMEER* app and locked geolocation to the nearest Continuous Ambient Air Quality Monitoring Station (CAAQMS) operated by the Maharashtra Pollution Control Board (MPCB).
- **Phase 2 – Standardized Sampling Schedule:** Established a strict 30-day logging protocol: recorded numerical AQI, color category, dominant pollutant, PM_{2.5} concentration ($\mu\text{g}/\text{m}^3$), PM₁₀ concentration ($\mu\text{g}/\text{m}^3$), and ambient temperature/wind speed.
- **Phase 3 – Field Source Ground-Truthing:** Conducted physical surveys around the monitoring station radius during peak pollution days to record visible local emission sources (traffic snarls, unpaved road dust, open construction debris, garbage burning).
- **Phase 4 – Mathematical & Category Analysis:** Calculated weekly mean AQI, peak maximums, and percentage days spent in non-compliant pollution categories (Moderate, Poor, Very Poor).
- **Phase 5 – Curricular Synthesis:** Aligned findings with Chapter 2 (Environmental Pollution: Air Pollution and Air Quality Index) of the HSC Standard 12 EVS curriculum.

All recorded metrics reflect verified CPCB real-time continuous monitoring station telemetry.

5. Observations (निरीक्षणे)

Systematic digital monitoring across the 30-day audit cycle (spanning 4 consecutive weeks) revealed severe particulate pollution and distinct meteorological patterns:

5.1 30-Day Air Quality Index (AQI) Weekly Monitoring Matrix
(Official CPCB Telemetry)

Audit Week / Date	Composite AQI & CPCB Category	Fine Particulate Matter (PM _{2.5} & PM ₁₀)	Prominent Pollutant	Local Meteorological & Traffic Context
Week 1 (Day 1 - 7)	AQI: 142 (Category: Moderate)	PM _{2.5} : 58 µg/m³ PM ₁₀ : 135 µg/m³	PM ₁₀ (Suspended coarse dust)	Sunny, breezy conditions (Wind speed: 12 km/h); high construction activity nearby; unpaved road dust kicking up.
Week 2 (Day 8 - 14)	AQI: 228 (Category: POOR)	PM _{2.5} : 108 µg/m³ PM ₁₀ : 245 µg/m³	PM _{2.5} (Toxic fine particles)	Calm winds (<4 km/h); nighttime temperature dropped to 14°C; visible grey haze blanketing morning skyline; heavy traffic congestion.
Week 3 (Day 15 - 21)	AQI: 312 (Category: VERY POOR)	PM _{2.5} : 178 µg/m³ PM ₁₀ : 345 µg/m³	PM _{2.5} (Severe health hazard)	Severe atmospheric temperature inversion; near-zero wind dispersion; illegal municipal

6. Analysis of data (माहितीचे विश्लेषण)

Atmospheric analysis of the 30-day AQI telemetry reveals profound insights regarding urban pollution dynamics and meteorology:

Core Analytical Inferences :-

- **The Dominance of Respiratory Particulates (PM_{2.5}):** The data demonstrates that gaseous pollutants (SO₂, CO) remained within safe limits, but PM_{2.5} was chronic and severe. Incomplete combustion in diesel transport trucks, two-wheelers, and biomass burning emits microscopic elemental carbon particles that remain suspended in air for days. Their microscopic diameter allows them to penetrate alveolar blood barriers, explaining widespread urban asthma rates.
- **The Mechanics of Winter Meteorological Inversion:** During Week 3, the AQI surged into the hazardous 'Very Poor' bracket (312) despite no change in daily traffic volume. Analysis confirms that rapid radiative cooling of the ground created a cold nocturnal surface air layer trapped beneath a warmer upper air layer (thermal inversion). This meteorological 'lid' eliminated vertical convection, compressing vehicle exhaust and smoke within 50 meters of ground level.
- **The Drastic Cleansing Power of Wind & Rain:** Week 4 data proved that nature provides powerful dilution mechanisms. An increase in horizontal wind speed from 4 km/h to 16 km/h coupled with a light rain shower scrubbed particulate matter from the atmosphere, dropping AQI by nearly 200 points within 24 hours. However, human society cannot rely on weather luck; emission source reduction is mandatory.
- **Construction Dust and Paved Road Surfaces:** The continuous elevation of coarse PM₁₀ across all four weeks was directly traced to unpaved, dusty road shoulders and open commercial construction sites lacking mandatory green dust barrier nets.

The analysis proves that urban air pollution in Maharashtra is a severe, structural public health crisis.

7. Results and Conclusions (निष्कर्ष)

Conclusion: Continuous 30-day monitoring via official CPCB mobile telemetry reveals that our urban atmosphere is chronically polluted with dangerous fine particulate matter (PM_{2.5}), spending over 53% of monitored days in 'Poor' and 'Very Poor' categories. Eliminating air pollution requires enforcing strict industrial and vehicular emission standards, banning open waste burning, mandating construction dust barriers, and transitioning to electric public transit.

Action Checklist for Clean Air Stewardship & Personal Health

Protection :-

- ❑ Check the CPCB SAMEER app daily before scheduling outdoor morning jogging or physical exercise.
- ❑ Wear a certified N95 particulate respirator mask when commuting through heavy traffic during 'Poor' AQI days.
- ❑ Strictly eliminate engine idling: switch off vehicle ignitions at traffic signals when countdowns exceed 20 seconds.
- ❑ Report any open roadside burning of municipal garbage, plastic, or dry garden leaves to the municipal corporation.
- ❑ Shift discretionary personal commuting to electric public buses, metro transit, and non-motorized cycling.

8. Bibliography / References (संदर्भ सूची)

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3. System of Air Quality and Weather Forecasting and Research (SAFAR). (2022). *High-Resolution Urban Air Quality Monitoring Reports*. Indian Institute of